Robin Leman

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EDUCATION

MSc in Computational Statistics and Machine Learning. UCL, London, UK SEP 2025 - SEP 2026

- **Coursework**: Probabilistic & Unsupervised Learning (Gatsby), Approximate Inference (Gatsby), Reinforcement Learning (DeepMind), Open-Endedness & General Intelligence, Bayesian Deep Learning, Generative AI.
- **Activities**: Head of ML Tutorials at the UCL AI Society; organized weekly workshops covering core ML concepts.

BSc in Physics and Computer Science. McGill University, Montreal, Canada SEP 2019 - MAY 2022

- **CGPA**: 3.82/4.00
- **Activities**: President of GameDev McGill. Led a 10-people team to organize hackathons & events for 100+ members.

RESEARCH EXPERIENCE

Causal Inference for Structured Outcomes (MSc Thesis, Gatsby Unit)
Supervisors: Houssam Zenati (Gatsby Unit), Arthur Gretton (Gatsby Unit / Google DeepMind)

- Training deep kernels to maximize the test power of counterfactual two-sample tests, enabling the detection of distributional treatment effects beyond mean shifts on structured outcomes like images, graphs or text sequences.

LLM-Guided Autocurricula via Semantic World Model
Supervisor: Ahmet H. Güzel (UCL Dark Lab)

- Developed LLM-driven autocurricula methods to generate targeted training scenarios from an agent's past learning trajectories in fully dreamed environments, improving sample efficiency over prioritized level replay.
- Trained a text-conditioned DiT world model using CLIP embeddings with ablation studies against the iMAC baseline.

Latent Dynamics Representations for Long-Horizon Consistency in World Models
Supervisor: Jagmohan Chauhan (UCL)

- Building a state-space JEPA world model to learn 3D scene representations from multiple camera viewpoints, improving long-range prediction accuracy over standard diffusion-based approaches.

WORK EXPERIENCE

Software Engineer. Electronic Arts, Vancouver, Canada JUL 2023 - SEP 2025

- Designed and implemented a high-performance Entity-Component-System simulation library, optimizing memory layout for cache-efficient data access in highly parallelized simulations.
- Optimized cross-platform networking for fast replication, interpolation and prediction, reducing memory usage by 50%.
- Developed an automated spec-test framework, enabling scalable automated validation and improving system resilience.

Associate Programmer. Relic Entertainment (SEGA), Vancouver, Canada JUN 2022 - JUN 2023

- Developed data-oriented real time simulation systems in an ECS architecture, optimizing data locality to maximize cache utilization.
- Implemented spatial partitioning and HFSM structures to accelerate nearest-neighbour queries on 1600+ entities.
- Optimized multi-threaded state tree data structures by 25%+, for highly performant AI systems.

Software Engineer Intern. Microsoft, Vancouver, Canada MAY 2021 - AUG 2021

- Profiled UE5's physics systems, running performance analysis & benchmarks to optimize simulation on Xbox & PC.

Generalist Programmer Intern. Ubisoft, Montreal, Canada JUN 2020 - AUG 2020

- Designed a voxel-based fluid simulation algorithm for real-time buoyancy within an ECS, data-oriented architecture.

PERSONAL PROJECTS

Cosal Built a platform for AI agents with tool orchestration, memory and safety layers, with Firecracker VM isolation.
Tiny-LLM Trained a 51M-parameter transformer language model, achieving perplexity 34 on WikiText-103.
Pico-Banana Built a 10M-parameter diffusion U-Net, using sinusoidal time embeddings, EMA and self-attention.
Cart-Pole RL Agents Implemented PPO from scratch in C++, reaching maximum reward in under 250 training batches.
GPU Raytracer Built a raytracer with CUDA/OpenGL interop and progressive rendering for real-time visual feedback.
Autocurricula Built an interview practice platform using LLMs to adaptively generate problems from user performance.

SKILLS

Python, PyTorch, C++, CUDA, Rust, C, Bash, C#, OpenGL, Unreal Engine, Unity, Git, Linux